

MA 321: Numerics of Differential Equations

Course Code	MA 321
Course Name	Numerics of Differential Equations
L-T-P-C	3-0-2-4
Prerequisites	IC253 Mutual Exclusion: MA 609
Intended for	B.Tech. 3rd year students
Distribution	Core for B. Tech. Mathematics and Computing, Elective for other disciplines.
Approval	59th BoA (Apr 2025)

Course Contents

- **Numerical Methods for Ordinary Differential Equations - Initial Value Problems (IVPs):** Taylor series method, Euler methods, Runge-Kutta methods; Multistep methods: Milne's method, Adams-Moulton method, System of equations, Higher order equations, Stiff equations. Boundary Value Problems (BVPs): Finite difference methods, Shooting method. [8 Lectures]
- **Introduction to PDEs:** - Classification of PDEs, Physical Classifications, Mathematical classifications, Introduction to Numerical Computing. [4 Lectures]
- **Introduction to Finite Difference approximations:** Simple method, General Method, Polynomial fitting. One-dimensional equations, Two-dimensional equations. Accuracy of Finite Difference Solutions: Consistency, Stability, Convergence analysis. [8 Lectures]
- **Finite Difference Methods for Hyperbolic Equations:** One-dimension and Two-dimension equations, Discretization, stability, CFL conditions, Explicit and Implicit schemes, Single-step and multi-step methods. [8 Lectures]
- **Finite Difference Methods for Parabolic Equations:** One-dimension and Two-dimension equations, Discretization, stability, Convergence of the scheme. Finite Difference methods for Convection-Diffusion Equations, sensitivity analysis. [9 Lectures]
- **Finite Difference Methods for Elliptic Equations:** One-dimension and Twodimension equations, Discretization, stability, Convergence of the scheme. [5 Lectures]

Laboratory Modules

The computer lab classes will complement the theoretical understanding of the students. All the numerical methods, discussed in the lecture classes, will be used to solve mathematical/physical problems through computer programming during the lab classes. This will provide a hands-on experience to the students on this subject.

Text Books

1. K. E. Atkinson, **An Introduction to Numerical Analysis**, 2nd Edition, John Wiley, 2008.
2. G. D. Smith, **Numerical Solutions to Partial Differential Equations**, 3rd Edition, Oxford University Press, 2004.

References

1. J. C. Strikwerda, **Finite Difference Schemes and Partial Differential Equations**, SIAM, 2004.
2. M. T. Heath, **Scientific Computing: An Introductory Survey**, McGraw Hill, 2002.